

Splunk Search Language (SPL) -Basic Detection Queries

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Overview

This report documents the development and validation of three Splunk Processing Language (SPL) queries designed to identify suspicious file creation, process execution, and network communication activity within Sysmon logs. The goal is to demonstrate basic detection engineering skills, noise reduction techniques, and investigative workflows used in a SOC environment.

I performed basic log analysis using Splunk Processing Language to identify suspicious process creation, network connections and suspicious file creation.

• Query 1: Suspicious file creation

```
index=main sourcetype=_json EventID=11
| eval Target=lower(TargetFilename)
| where (like(Target, "%\\temp\\%") AND (like(Target, "%\\.exe") OR like(Target, "%\\.dll")))
| where NOT like(Target, "%\\program files\\%")
| where NOT like(Target, "%\\windows\\%")
| table _time, user, computer, TargetFilename, Image, ProcessID
| sort _time
```

The screenshot displays the Splunk Search interface. At the top, there's a navigation bar with 'Search & Reporting' highlighted. Below it, the 'New Search' section shows the input query: `index=main sourcetype=_json EventID=11 | eval Target=lower(TargetFilename) | where (like(Target, "%\\temp\\%") AND (like(Target, "%\\.exe") OR like(Target, "%\\.dll"))) | where NOT like(Target, "%\\program files\\%") | where NOT like(Target, "%\\windows\\%") | table _time, user, computer, TargetFilename, Image, ProcessID | sort _time`. The search results show 2,427 events. The table below lists the first few results.

_time	user	computer	TargetFilename	Image	ProcessID
2025-12-10 04:06:08		MSEDGEWIN10	C:\Users\IEUser\AppData\Local\Temp_MEI72882\python27.dll	C:\windows\temp\supply.exe	
2025-12-10 04:06:08		MSEDGEWIN10	C:\Users\IEUser\AppData\Local\Temp_MEI72882\msvcr90.dll	C:\windows\temp\supply.exe	
2025-12-10 04:06:08		MSEDGEWIN10	C:\Users\IEUser\AppData\Local\Temp_MEI72882\msvcp90.dll	C:\windows\temp\supply.exe	
2025-12-10 04:06:08		MSEDGEWIN10	C:\Users\IEUser\AppData\Local\Temp_MEI72882\msvcr90.dll	C:\windows\temp\supply.exe	

Explanation:

Line 1 searches the main index and filter for EventID = 11, which is sysmon File Create ID

Line 2 creates a new field called Target and converts the existing TargetFilename field to lowercase to avoid missing matches due to capitalization.

Line 3 filters file that was created in the Temp directory and ends with either .exe or .dll

Line 4 and 5 reduces noise by excluding benign paths from the search.

Line 6 outputs the results of the search in a table form specifying the necessary fields

Line 7 sorts the outputs by time of events.

● Query 2: Suspicious Process Creation

```
index=main sourcetype=_json EventID=1 Image IN ("powershell.exe", "cmd.exe",
"wscript.exe", "cscript.exe")
| eval PI = lower(ParentImage), PC=lower(ParentCommandLine)
| eval score=0
| eval score=score + if(like(PI,"%\\Downloads\\%") OR like(PI,"%\\temp\\%"),1,0)
| eval score=score + if(like(PC,"%powershell%") OR like(PC,"%cmd.exe%") OR
like(PC,"%wscript%") OR like(PC,"%cscript%"),1,0)
| where score >= 1
| table _time, host, user, Image, ParentImage, ProcessId, ParentProcessId,
ParentCommandLine, score
| sort _time
```

← → ↻ http://127.0.0.1:8000/en-US/app/search/search?q=search index%3Dmain sourcetype%3D_json EventID%3D1 Sign in

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New Search

Save As Create Table View Close

```

index=main sourcetype=_json EventID=1 Image IN ("powershell.exe", "cmd.exe", "wscript.exe", "cscript.exe")
| eval PI = lower(ParentImage), PC=lower(ParentCommandLine)
| eval score=0
| eval score=score + if(like(PI,"%\\Downloads\\%") OR like(PI,"%\\temp\\%"),1,0)
| eval score=score + if(like(PC,"%powershell%") OR like(PC,"%cmd.exe%") OR like(PC,"%wscript%") OR like(PC,"%cscript%"),1,0)
| where score >= 1
| table _time, host, user, Image, ParentImage, ProcessId, ParentProcessId, ParentCommandLine, score
| sort _time

```

Time range: All time

✓ 21 events (before 1/12/26 5:54:59.000 AM) No Event Sampling Job

Events (21) Patterns **Statistics (21)** Visualization

Show: 100 Per Page Format Preview: On

_time	host	user	Image	ParentImage	ProcessId	ParentProcessId
2025-12-10 04:06:08	Brown		C:\Windows\SysWOW64\cmd.exe	C:\Windows\SysWOW64\WindowsPowerShell\v1.0\powershell.exe	9628	
2025-12-10 04:06:08	Brown		C:\Windows\SysWOW64\cmd.exe	C:\Windows\SysWOW64\WindowsPowerShell\v1.0\powershell.exe	6172	

Explanation:

Line 1 searches the main index, filter for event ID=1; which is sysmon process creation ID. Then search the Image field for processes powershell,cmd,wscript and cscript. Which are common scripting and commandline tools mostly abused by attackers.

Line 2 creates new fields PI and PC which convert Parent image and ParentcommandLine field to lowercase respectively. This ensures that the search strings are case insensitive.

Line3 creates a new numeric field “score=0” which starts all events at zero. It will be used as a simple risk-scoring mechanism

Line4, add one point if the parent process came from a Downloads folder OR Temp directory. Otherwise, add 0

Line5, add one point if the parent command Line field contains keywords powershell,cmd.exe,wscript or cscript. Otherwise, add 0.

Line 6 filter by risk score i.e keep events that score more at least 1. That means one suspicious behaviour was detected.

Line 7 displays the result in table format with relevant fields.

Line 8 sorts the results by time.

• Query 3: Network Connections By IP

```

index=main sourcetype=_json EventID=3
| stats count by destinationIp
| sort - count

```

